

```
BFS.c - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
<global> BFS(int adj[100][100], int n, int start) : void
Start here x BFS.c x
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  #define MAX 100
5  int queue[MAX], front=-1, rear=-1;
6  int visited[MAX];
7
8  void enqueue(int v) {
9      if (rear==MAX-1)
10         printf("Queue Overflow\n");
11     else {
12         if (front==MAX-1)
13             front=0;
14         rear++;
15         queue[rear]=v;
16     }
17 }
18 int dequeue() {
19     int v;
20     if (front==MAX-1 || front>rear)
21         return -1;
22     else {
23         v=queue[front];
24         front++;
25         return v;
26     }
27 }
28 void BFS(int adj[MAX][MAX], int n, int start) {
29     int i, current;
30     for (i=0; i<n; i++)
31         visited[i]=0;
32     enqueue(start);
33     visited[start]=1;
34     printf("BFS Traversal:");
35     while ((current=dequeue())!=-1) {
36         printf("%d", current);
37         for (i=0; i<n; i++) {
38             if (adj[current][i]==1 && !visited[i]) {
39                 enqueue(i);
40                 visited[i]=1;
41             }
42         }
43     }
44     printf("\n");
45 }
46 int main() {
47     int n, i, j, start;
48     int adj[MAX][MAX];
49     printf("Enter number of vertices:");
50     scanf("%d", &n);
```

C:\Users\BMSCECE-L4\Desktop\1BM24CS173\BFS.c

C/C++

Windows (CR+LF)

WINDOWS-1252

Line 44, Col 18, Pos 905

Insert

Read/Write

default

ABX

```
BFS.c - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
<global> BFS(int adj[100][100], int n, int start) : void
Start here x BFS.c x
11     else{
12         if(front==0)
13             front=0;
14             rear++;
15             queue[rear]=v;
16     }
17 }
18 int dequeue(){
19     int v;
20     if(front==0 || front>rear)
21         return -1;
22     else{
23         v=queue[front];
24         front++;
25         return v;
26     }
27 }
28 void BFS(int adj[MAX][MAX],int n,int start){
29     int i,current;
30     for(i=0;i<n;i++){
31         visited[i]=0;
32         enqueue(start);
33         visited[start]=1;
34         printf("BFS Traversal:");
35         while((current=dequeue())!=-1){
36             printf("%d",current);
37             for(i=0;i<n;i++){
38                 if(adj[current][i]==1&&!visited[i]){
39                     enqueue(i);
40                     visited[i]=1;
41                 }
42             }
43         }
44         printf("\n");
45     }
46 }
47 int main(){
48     int n,i,j,start;
49     int adj[MAX][MAX];
50     printf("Enter number of vertices:");
51     scanf("%d",&n);
52     printf("Enter adjacency matrix of the graph:\n");
53     for(i=0;i<n;i++){
54         for(j=0;j<n;j++){
55             scanf("%d",&adj[i][j]);
56         }
57     }
58     printf("Enter starting vertex for BFS:");
59     scanf("%d",&start);
60     BFS(adj,n,start);
61     return 0;
62 }
```

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C/C++

Windows (CR+LF)

WINDOWS-1252

Line 44, Col 18, Pos 905

Insert

Read/Write

default

ABX

Enter number of vertices:6

Enter adjacency matrix of the graph:

```
0 1 1 0 0 0
1 0 0 0 0 1
1 0 0 1 0 1
0 0 1 0 1 1
0 0 0 1 0 1
0 1 1 1 1 0
```

Enter starting vertex for BFS:0

BFS Traversal:0 1 2 5 3 4

Process returned 0 (0x0) execution time : 13.279 s

Press any key to continue.